

Support Vector Machine (SVM)

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1 Introduction

Support Vector Machine (SVM) is one of the most powerful and widely used supervised machine learning algorithms for classification and regression tasks. It was introduced by Vladimir Vapnik and his colleagues in the early 1990s and has become one of the most successful algorithms in statistical learning theory.

Although SVM can be applied to both classification and regression problems, it is mainly known for binary classification, where the objective is to separate data into two different classes.

Unlike many classification algorithms that simply search for a boundary between classes, SVM seeks to find the **optimal separating boundary**. The optimal boundary is the one that maximizes the distance between the two classes. This distance is called the **margin**. By maximizing the margin, SVM usually produces a classifier that generalizes better to unseen data.

Another important characteristic of SVM is that only a small number of training samples determine the final decision boundary. These critical samples are called **support vectors**. All other training samples have little or no influence on the position of the separating hyperplane.

SVM is particularly effective when the dataset has many features or dimensions. It also performs well even when the number of features is greater than the number of training samples. Furthermore, by using kernel functions, SVM can solve both linear and nonlinear classification problems.

Because of its high accuracy and strong theoretical foundation, SVM is widely used in many real-world applications, including image recognition, medical diagnosis, handwriting recognition, spam detection, text classification, and bioinformatics.

2 Basic Idea of Support Vector Machine

The primary objective of a Support Vector Machine is to separate two classes of data by constructing the best possible decision boundary.

Suppose a dataset contains two classes represented by circles and squares. There are often many possible lines that can separate these two classes correctly. For example,

○ ○ ○

Several separating lines can correctly classify these samples. However, not all separating lines are equally good.

A line passing very close to one class may classify the training data correctly but can easily misclassify new observations. Such a classifier has poor generalization.

Instead of choosing any separating line, SVM searches for the line that leaves the largest possible distance between the two classes.

This distance is called the **margin**.

The larger the margin, the more robust the classifier becomes against noise and small variations in the data.

Therefore, the main idea of SVM can be summarized as follows:

Find the separating hyperplane that maximizes the margin between the two classes.

This principle is known as the **Maximum Margin Principle**.

Figure 1 illustrates the concept.

The central solid line represents the optimal hyperplane, while the dashed lines indicate the margins on both sides. The training samples that lie on these margins are called the support vectors because they determine the position of the hyperplane.

Before studying the mathematical formulation of SVM, it is important to understand the concepts of the hyperplane, margin, and support vectors, which form the foundation of the algorithm.

3 Hyperplane

A **hyperplane** is the decision boundary used by a Support Vector Machine (SVM) to separate different classes of data.

In a two-dimensional space, the hyperplane is a straight line. In a three-dimensional space, it is a plane. In higher-dimensional spaces, it is called a **hyperplane**.

The objective of SVM is to determine the optimal hyperplane that best separates the classes while maximizing the margin between them.

3.1 Hyperplane in Two Dimensions

Suppose we have two features, x_1 and x_2 . A hyperplane is simply a straight line that divides the data into two regions.

For example,

$$x_1 + x_2 = 6$$

is a straight line that separates the feature space into two parts.
Any point located on this line satisfies

$$x_1 + x_2 = 6.$$

Points above the line belong to one region, while points below the line belong to the other region.

3.2 General Equation of a Hyperplane

For a dataset with n features, the equation of a hyperplane is

$$w^T x + b = 0$$

where

- $x = [x_1, x_2, \dots, x_n]^T$ is the feature vector,
- $w = [w_1, w_2, \dots, w_n]^T$ is the weight vector,
- b is the bias (intercept),
- $w^T x$ represents the dot product between w and x .

Expanding the dot product gives

$$w^T x = w_1 x_1 + w_2 x_2 + \dots + w_n x_n.$$

Therefore, the hyperplane equation can also be written as

$$w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b = 0.$$

3.3 Meaning of the Weight Vector

The vector w determines the orientation (direction) of the hyperplane.

Changing the values of w rotates the hyperplane.

For example,

$$2x_1 + x_2 - 5 = 0$$

and

$$x_1 + 3x_2 - 5 = 0$$

represent two different hyperplanes because their weight vectors are different.

3.4 Meaning of the Bias

The parameter b shifts the hyperplane without changing its orientation.

For example,

$$x_1 + x_2 - 4 = 0$$

and

$$x_1 + x_2 - 8 = 0$$

have the same orientation because the weight vector is identical, but the second hyperplane is translated to a different position.

3.5 Decision Function

The decision function of an SVM is

$$f(x) = w^T x + b.$$

This function determines on which side of the hyperplane a sample lies.

The classification rule is

$$\{ f(x) > 0 \Rightarrow \text{Class}+1, f(x) < 0 \Rightarrow \text{Class}-1, f(x) = 0 \Rightarrow \text{Point lies on the hyperplane.} \}$$

Thus, the sign of the decision function determines the predicted class.

3.6 Simple Example

Consider the hyperplane

$$x_1 + x_2 - 6 = 0.$$

The corresponding decision function is

$$f(x) = x_1 + x_2 - 6.$$

Consider the point

$$P = (2, 2).$$

Substituting into the decision function,

$$f(P) = 2 + 2 - 6 = -2.$$

Since

$$-2 < 0,$$

the point belongs to Class -1 .
Now consider another point

$$Q = (5, 4).$$

Then,

$$f(Q) = 5 + 4 - 6 = 3.$$

Since

$$3 > 0,$$

the point belongs to Class $+1$.

Therefore, the hyperplane successfully separates the two points into different classes.

3.7 Summary

The hyperplane is the decision boundary that separates different classes in an SVM model.

Its equation is

$$w^T x + b = 0,$$

where the weight vector w controls the orientation of the hyperplane and the bias b controls its position.

The sign of the decision function

$$f(x) = w^T x + b$$

determines the predicted class of every observation.

In the next section, we will study the concept of the **margin**, which explains why SVM chooses one hyperplane instead of another.

3.8 Numerical Example

Consider the following hyperplane:

$$x_1 + x_2 - 6 = 0.$$

The corresponding decision function is

$$f(x) = x_1 + x_2 - 6.$$

The classification rule is

$$\{ f(x) > 0 \Rightarrow \text{Class} + 1, f(x) < 0 \Rightarrow \text{Class} - 1. \}$$

Suppose we have the following dataset.

Step 1: Classify Point A

For Point A,

$$x = (1, 1).$$

Compute the decision function:

$$f(A) = 1 + 1 - 6 = -4.$$

Since

$$-4 < 0,$$

Point A is classified as Class -1 .

Step 2: Classify Point B

For Point B,

$$x = (2, 2).$$

$$f(B) = 2 + 2 - 6 = -2.$$

Since

$$-2 < 0,$$

Point B is classified as Class -1 .

Step 3: Classify Point C

For Point C,

$$x = (5, 4).$$

$$f(C) = 5 + 4 - 6 = 3.$$

Since

$$3 > 0,$$

Point C is classified as Class $+1$.

Step 4: Classify Point D

For Point D,

$$x = (6, 5).$$

$$f(D) = 6 + 5 - 6 = 5.$$

Since

$$5 > 0,$$

Point D is classified as Class +1.

Summary of Results

All four samples are classified correctly by the hyperplane.

Therefore, the hyperplane

$$x_1 + x_2 - 6 = 0$$

successfully separates the two classes.

At this stage, we have learned how a hyperplane classifies data points. In the next section, we will see that SVM does not choose just any separating hyperplane. Instead, it selects the one that maximizes the distance between the two classes, known as the **maximum margin**.

[scale=0.8]
[blue] (1,1) circle (2.5pt); [blue] (2,2) circle (2.5pt); [blue] (2,0.5) circle (2.5pt);
[blue] (3,1.5) circle (2.5pt);
[red] (7,5) rectangle +(0.15,0.15); [red] (8,6) rectangle +(0.15,0.15); [red]
(7,6.5) rectangle +(0.15,0.15); [red] (8.5,5.2) rectangle +(0.15,0.15);
[thick] (4.5,-0.5)–(4.5,7);
[dashed] (3.7,-0.5)–(3.7,7); [dashed] (5.3,-0.5)–(5.3,7);
at (4.9,6.6)Hyperplane; at (3.2,6.2)Margin; at (5.8,6.2)Margin;

Figure 1: Maximum-margin separating hyperplane.

Point	Coordinates (x_1, x_2)	Actual Class
A	(1,1)	-1
B	(2,2)	-1
C	(5,4)	+1
D	(6,5)	+1

Table 1: Training dataset

Point	Calculation	$f(x)$	Predicted Class
A	$1 + 1 - 6$	-4	-1
B	$2 + 2 - 6$	-2	-1
C	$5 + 4 - 6$	3	+1
D	$6 + 5 - 6$	5	+1

Table 2: Classification results